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AlphaFAX®
User's Guide

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CHAPTER 1

INTRODUCTION TO AlphaFAX

What is AlphaFAX?

AlphaFAX uses a fax modem to send a facsimile message (fax) from your computer to a remote facsimile machine. It also allows you to receive faxes from a fax machine or another AlphaFAX user. AlphaFAX gives full fax access to every user of your computer and, in a networked environment, to every user of the network. You can now send a fax message to a fax machine anywhere in the world without even leaving your desk.

Some of AlphaFAX's features include:

- Activity log tells you what faxes have been sent and received, and when.
- You can preview a fax before it is sent by printing it on a local printer.
- You can send a single fax to multiple fax machines by specifying a distribution list containing a list of fax phone numbers.
- You can easily include text and bitmap images in your faxes.
- Receipt of faxes is automatic, with no operator intervention required.
- You can select a specific time to send a fax.
- Fax machine phone numbers can be defined and specified in your fax as a symbolic name such as "Capistrano Depot."

How Do I Use AlphaFAX?

The first step in sending a fax is to create a text file using one of the word processing programs supported. Next, within the file type the phone number of the fax machine you want to send the message to, and print it (selecting the "fax printer"). Instead of appearing on a printer, the file is converted by AlphaFAX to a fax format and sent to the specified fax machine. In effect, the other fax machine becomes a printer on your computer.

Installation

The AlphaFAX product consists of a fax modem and software programs. For information on installing these components, see the *AlphaFAX Installation Guide*. That document also discusses hardware and software requirements.

About this Book

This manual conforms to other Alpha Micro publications in its use of a standard set of graphics conventions. We hope these graphics simplify our examples and make them easier for you to use.

SYMBOL	MEANING
TEXT	Bold text in an example of user/computer communication represents the characters you type. When the text is all capitals, type exactly what's shown. When it is lower case, it represents a variable part of a command, such as a file name.
TEXT	Text in this type face is used for: characters the computer displays on your screen, program examples, and command syntax. When showing command syntax, all capitals indicates a constant part of the command line, such as the command name, while lower case indicates indicates variable information, such as a file name.
text	Bold italic text in an explanation of command use indicates a parameter accepted by that command. For example: "The FAX command accepts the ***pic*** argument, which selects the Product Installation Code."
KEY	In our examples, the keycap symbol appears whenever you need to press a certain key on your terminal keyboard. The name of the key is inside the keycap, like this: ENTER.
	This symbol means "halt!" It indicates an important note you should read carefully before going further in the documentation. Usually, text next to this symbol contains instructions for something you MUST or MUST NOT do, so read it carefully.
	This symbol means "hint." It indicates a helpful bit of information, or a "short cut" that could save you time or trouble.
	This symbol means "remember." It indicates something you should keep in mind while you are following a set of instructions.

CHAPTER 2

PREPARING THE DOCUMENT TO BE FAXED

Introduction

Use AlphaWRITE, AlphaVUE, or SuperVUE to create the message you want to send. A fax message is no different from any other text file except that you include special commands in the text that direct AlphaFAX where and how to send the fax. Briefly, here is what you have to do to convert a text file into a fax message:

- Use the @@TELNO command in the document to tell AlphaFAX the phone number of the fax machine you want to send the fax to.
- If you want to print a copy of the fax on your local printer before sending it off, use the @@PREVIEW command in your file.

That's all there is to it. Print the document to the fax spooler and the fax will be sent. Your application program could just as easily add the AlphaFAX commands to fully automate the process.

What's Next?

This chapter gives more details on using the @@TELNO and @@PREVIEW commands. The next chapter tells you how to send, receive, and print faxes. Chapter 4 tells you about the other AlphaFAX commands you can use to make sending faxes even more convenient. Using those commands, you will be able to:

- Defer sending the fax until a specified time
- Include text or bitmap images in the fax
- Distribute a single fax to a number of different fax machines
- Specify an ID code to be used in the AlphaFAX activity log to refer to the fax
- Select high or low resolution in which to send the fax

Creating the Fax

When you use a word processing program to create the fax document, you must make sure that each page of your fax message will fit comfortably on the page when it is sent, or it will not appear correctly on the remote fax machine.

Remember that most of the world uses a different paper size— called A4— rather than the US standard letter size (8.5" x 11"). A4 is slightly narrower and longer than letter size. Since you cannot be sure what size paper is loaded in the remote fax machine, it is best to format your documents so that they will appear correctly on any paper size. Keeping your fax within an 8" x 11" area fits virtually any size paper.

AlphaFAX includes support for bold, italic, and underscored text. These are accessed from within your word processor just as with any other printer. AlphaFAX also supports proportionally spaced text which can be accessed from AlphaWRITE 2.1 and later.

Here are some specific settings for your AlphaWRITE document header that will ensure a fax prepared using AlphaWRITE will fit either 8.5" x 11" paper or A4 paper.

For AlphaWRITE 1.2A, Set:

Page Length: 66
Text Length: 57
Space Before Footnote: 0

and, remember to use the PAGINATE command to format the fax into pages of the correct length.

For AlphaWRITE 2.0 and Later, Set:

Top Margin: 0
Title Length: 0
Page Length: 11"
Text Length: 9"
Space Before Footnote: 0
Footer Length: 0

For more information on using AlphaWRITE to create faxes, see Chapter 7, "Using AlphaFAX with AlphaWRITE."

AlphaFAX Command Format

AlphaFAX commands appear in your text file preceded by the @@ symbols; for example:

```
@@TELNO=1-714 957 8500
```

tells AlphaFAX what phone number to send the fax to. AlphaFAX commands must appear at the beginning of a line.

Specifying the Fax Machine Telephone Number

To send a fax, you must include the telephone number of the destination fax machine in the message file. This is done via the @@TELNO command. The format of this command is:

```
@@TELNO=dialstring
```

dialstring is a series of digits to be dialed. Do not include any outside access codes (such as a 9), as this will be included automatically. If needed, you can include the characters * and #, as well as a comma (,) to insert a brief pause, and W to wait for a dialtone. You may also include spaces, hyphens, and parentheses symbols, all of which are ignored.

Remember that you must include all codes needed to dial the number except outside access codes. Don't forget to include a 1 before the area code when dialing long distance.

The dialstring may also be a character string which refers to an entry in the FAXLST.DAT phone list. See Chapter 4 for more details.

Using the AlphaWRITE Glossary for "Speed-dialing"

The AlphaWRITE glossary can be used as a "speed dialer" for the fax machine phone number. Enter a fax phone number into the glossary under a two-digit code, and then you may enter it into the fax document with a simple key sequence.

Previewing the Fax Before Sending It

Once you have finished your fax, you may want to take a look at it before sending it. If your fax contains the @@PREVIEW command, AlphaFAX sends a copy of it to a local printer for previewing or documentation purposes. Including the @@PREVIEW command in a fax document forces the @@TELNO command to be ignored. The format of the command is:

```
@@PREVIEW
```

CHAPTER 3

SENDING, RECEIVING AND PRINTING FAXES

Getting Help

This chapter contains information on sending, receiving and printing faxes. If you have any problems with any of these activities (for example, if a fax is not successfully sent), please see "Troubleshooting AlphaFAX" in the *AlphaFAX Installation Guide* for help in troubleshooting the problem.

Sending a Fax

Once your message file is complete and contains the telephone number of the fax machine you want to send the fax to, as discussed in the previous chapter, sending the fax is a simple matter of printing the file to the fax printer. From within AlphaWRITE, or other programs that provide their own interface to the system printer spooler, you see the fax printer listed as one of your printer choices. (Extra prompting information may be included in the printer list, such as "Send a fax.") Choose the fax printer, and print your message.

From AMOS command level, you can send a fax by using the appropriate form of the "print" command for your computer, and specifying the fax printer. For example:

PRNT FAX=MYFAX.LST RETURN

or:

PRINT FAX=MYFAX.LST RETURN

A copy of your outgoing fax is placed in FAXOUT:, where AlphaFAX transmits it. After the fax is successfully sent, AlphaFAX deletes it from FAXOUT:. Your original document file remains where you created it. (Of course, you can erase the source file yourself, if you want to.)

Cancelling a Fax Transmission

Sometimes it becomes necessary to tell AlphaFAX to stop trying to send a fax. For example, when reviewing the fax activity log (discussed in Chapter 5), you might see that a particular fax has been repeatedly transmitted without success (which might be because the telephone number for the fax machine is incorrect). In such a case, you will want to tell AlphaFAX to stop trying to send the fax.

To cancel the transmission of a fax, use the DELFAX command. The format of the DELFAX command is:

```
DELFAX fax-id
```

fax-id is the ID of the fax you want to remove from the transmission queue. For example:

```
DELFAX Sally RETURN
```

The fax ID is either the one you specified in the fax document via the @@ID command, or the default one AlphaFAX assigned if no @@ID command was in the fax document. (To see what a fax's ID is, look at the activity log. Each entry in the log lists the ID of the fax whose status is being logged.)

If multiple faxes exist with the same ID, DELFAX cancels the one listed in the log with the highest number of transmission retries.

Receiving a Fax

The receipt of incoming faxes is entirely automatic, and requires no operator intervention. When an incoming call is detected, the software automatically answers the phone, receives the fax (placing it on disk), notifies the operator a fax was received, and optionally prints the document. (To request that AlphaFAX automatically print all incoming faxes, you will want to modify the PRINT command in the FAX.INI AlphaFAX initialization file—see the *AlphaFAX Installation Guide* for details.)

All incoming faxes are placed in the ersatz device FAXIN:.

Printing a Received Fax

Once an incoming fax has been received, it can be printed at any time by using the PRTFAX command. The PRTFAX command format is:

```
PRTFAX filespec
```

filespec specifies the fax document to be printed.

CHAPTER 4

ADDITIONAL AlphaFAX FEATURES

Introduction

In the previous chapters, you saw how simple it is to send and receive faxes using AlphaFAX. This chapter contains information on the additional AlphaFAX features that make sending and receiving faxes even more convenient. We talk about:

- How to define names for the fax machine telephone numbers you use the most.
- How to define a distribution list so you can send a single fax to multiple fax machines.
- How to include images and text in a fax.
- How to select a specific time to send the fax.
- How to define an ID code that the fax activity logs use when referring to your fax.
- How to generate multiple faxes from a single text file.

Sending to Special Telephone Numbers

AlphaFAX normally validates the telephone number you specify in a @@TELNO command before dialing to make sure that it is a valid number. AlphaFAX checks for the correct number of digits, makes sure that a 1 is sent prior to an area code, and checks to make sure that no area code is sent if it is a local call. AlphaFAX attempts to correct such errors when it can.

Because AlphaFAX validates the telephone number against the rules defined by the North American Dialing Plan, international users may wish to disable this feature. North American users may also wish to disable the feature when dialing internal PBX numbers or including special access codes.

There are two ways to disable AlphaFAX's validation feature. To completely turn off the feature, modify the FAX.INI file to include the line:

```
VALIDATE=NO
```

To disable the feature on a fax-by-fax basis, enclose the dialstring argument of the @@TELNO command in quotes. This causes AlphaFAX to dial the number exactly as specified, without any validation. For example, the following command causes AlphaFAX to dial exactly what is in quotes, even though it is not a valid number by AlphaFAX's rules:

```
@@TELNO="544463770"
```

Defining Names for Telephone Numbers

AlphaFAX allows you to redefine telephone numbers as names to make them easier to remember, by entering your definitions in a file named FAXLST.DAT.

To create a FAXLST.DAT file, make a copy of the FAXLST.EG sample file included with AlphaFAX, and use AlphaVUE to edit and modify it:

```
LOG DSK0:[7,0] RETURN
COPY FAXLST.DAT=FAXLST.EG RETURN
VUE FAXLST.DAT RETURN
```

The FAXLST.DAT file contains a `name=number` line for each telephone number. ***name*** is the name that will be used on the @@TELNO line of the fax. ***number*** contains the telephone number of the remote fax machine. Do not put spaces immediately before or after the equal (=) sign. Spaces within the name or within the telephone number are allowed. You may optionally add a comment to any line by preceding the comment with a semicolon (;). For example:

```
; Company Offices
RGS=1-303 555 8422  ; Ridgway office
LGB=1-714 555 2402  ; Main depot
```

Once FAXLST.DAT has been modified for your needs, you can use any of the names defined in it in the @@TELNO command in your faxes.

Defining a Distribution List

Sending a single fax document to a list of fax machines is made possible through the @@DISTRIBUTE command, which you place in your fax in place of the @@TELNO command. The format of this command is:

```
@@DISTRIBUTE=filespec
```

filespec specifies a text file that is a list of either phone numbers or names in the FAXLST.DAT file. (See "Defining Names for Telephone Numbers," above for information on FAXLST.DAT.)

For example, within the document you wish to distribute to multiple sites, you might have the command:

```
@@DISTRIBUTE=DSK3:DIST.DAT[30,1]
```

Within the file DIST.DAT, place the list of fax numbers you wish the fax to be sent to. For example:

```
1 714 957-8705
LGB
MAIN OFFICE
387-2233
```

Remember that any names you specify in the distribution list must be defined in the FAXLST.DAT file.

Because the actual processing of the distribution list is done by the printer spooler— which is not logged into the same disk and account that you are— it is necessary to include a complete file specification for the distribution list, including device, filename and account. For network applications you can optionally include a CPU specification or ersatz device.

Including Images in the Fax

You have the option of including bitmap images in your document via the @@IMAGE command. Both BMP (Alpha Micro format bitmap) and PCX format images are supported. The format of this command, which you place in your fax document, is:

```
@@IMAGE=filespec{;position}
```

filespec specifies a bitmap image file that is to be placed in the outgoing document. The default extension is .BMP. The ***position*** argument is optional and allows you to specify the horizontal placement of the image. If you leave off this argument, the image will be placed flush against the left edge of the paper. Valid arguments for position include:

- R Places the image flush against the right edge of the paper.
- C Centers the image on the paper.
- L Places the image flush against the left edge of the paper.
- n *n* is a numeric value, specifying the number of inches from the left edge of the paper that you wish the image indented.

AlphaFAX supports both BMP and PCX format images. BMP images are stored in Alpha Micro's bitmap image format and are stored with the extension .BMP. BMP format is fully supported by AMIGOS, the Alpha Micro Graphics Operating System. PCX images are stored in a special bitmap format that is commonly supported on Macintosh and PC-compatible computer systems, allowing preparation of images on such computers for later use with AlphaFAX. AlphaFAX supports only monochrome (single-bit) PCX images.

The default extension for the @@IMAGE command is .BMP. If you wish to specify a PCX format file, you *must* specify an extension of .PCX.

Because the actual processing of the image file is done by the printer spooler— which is not logged into the same disk and account that you are— it is necessary to include a complete file specification for the bitmap image file including device, filename and account. For network applications you can optionally include a CPU specification or ersatz device.

Including Text in the Fax

You also have the option of including text in your document via the @@INCLUDE command. The format of this command, which you type as part of your fax document, is:

```
@@INCLUDE=filespec
```

filespec specifies a text file that is to be placed in the outgoing document. A common use for this command is for faxing reports or other documents generated by computer programs (for example, a monthly inventory report).

Because the actual processing of the file containing the included text is done by the printer spooler— which is not logged into the same disk and account that you are— it is necessary to include a complete file specification for the text file to include, including device, filename and account. For network applications you can optionally include a CPU specification or ersatz device.

Overlaying Images onto the Fax

When you include an image using the @@IMAGE command, it places the image onto the paper *instead* of text. While ideal for letterhead and signatures, it does not allow you to place images and text on the same line.

The @@OVERLAY command allows you to include an image on the page which the text is then printed on top of. This allows you to use an image of a pre-printed form and place text over it to make it appear as if the form had been typed.

The @@OVERLAY command supports both BMP and PCX format images.

The format of this command, which you place in your fax document, is:

```
@@OVERLAY=filespec
```

filespec specifies a bitmap image file that is to be placed in the outgoing document. The default extension is .BMP.

AlphaFAX supports both BMP and PCX format images. BMP images are stored in Alpha Micro's bitmap image format and are stored with the extension .BMP. BMP format is fully supported by AMIGOS, the Alpha Micro Graphics Operating System. PCX images are stored in a special bitmap format that is commonly supported on Macintosh and PC-compatible computer systems, allowing preparation of images on such computers for later use with AlphaFAX. AlphaFAX supports only monochrome (single bit) PCX images.

The default extension for the @@OVERLAY command is .BMP. If you wish to specify a PCX format file, you *must* specify an extension of .PCX.

Only one @@OVERLAY command can be specified for a single page; any additional @@OVERLAY commands for the same page are ignored.

The @@OVERLAY command specifies an image to be placed on the current page. If you wish to use the same overlay on all pages of your fax, use the @@OVERLAYALL command, which uses the same command format as the @@OVERLAY command. The @@OVERLAYALL command causes the current page and all subsequent pages to use the specified overlay file.

To create a fax which uses one overlay on the first page, and a different overlay on all subsequent pages, place a @@OVERLAY command on the first page, specifying the overlay to be used by that page. Immediately follow it with a @@OVERLAYALL command which specifies the overlay for all subsequent pages.

Because the actual processing of the image file is done by the printer spooler— which is not logged into the same disk and account that you are— it is necessary to include a complete file specification for the bitmap image file, including device, filename and account. For network applications you can optionally include a CPU specification or ersatz device.

Selecting Fax Image Resolution

Facsimile transmission is normally done at a "standard" resolution of 100 dots-per-inch along the vertical axis of the page. You may select the "fine" mode of 200 dots-per-inch by including the @@FINE command in your fax document.

Currently, selecting fine mode transmission is only an advantage when images are included in the document being sent. If no images are included, avoiding the use of fine mode reduces transmission time and phone charges.

The @@FINE command must precede all printable text within your document. If you have text prior to the @@FINE command the document will be sent at standard resolution.

Selecting a Specific Time to Send the Fax

The transmission of a fax may be delayed up to 24 hours via the @@DEFER command. The format of this command, which you place in your fax document, is:

```
@@DEFER=time
```

time indicates a specific time to begin processing the FAX transmission. The time specified is always considered to be in the future and a delay of up to 24 hours maximum may be specified. You may specify the time in either 12 hour or 24 hour format. If you use 12-hour format, include AM or PM in the time. For example:

```
@@DEFER=3:45 AM
```

Defining a Fax ID

AlphaFAX automatically generates fax logs that give information on faxes sent and received. The fax logs identify your faxes by a date code unless you specify an ID via the @@ID command. The format of the command, which you place in your fax document, is:

```
@@ID=string
```

string is a character string of up to 10 characters. For example:

```
@@ID=Sally 3
```

It is a good idea to include a @@ID command in all of your faxes. This allows you to easily spot your fax when others are also using AlphaFAX. To make it easier to identify individual faxes, it is a good idea to give each fax you send a unique ID. Your system operator may set AlphaFAX to reject any fax that does not contain a @@ID command.

Requesting Notification of Fax Transmission

You can optionally request that AlphaFAX notify you when a document has been transmitted via the @@NOTIFY command. AlphaFAX does this by sending a message to the specified job advising you of the successful transmission of a document, or of the fact that AlphaFAX has given up trying to transmit the document. The format of the command, which you place in your fax document, is:

```
@@NOTIFY=job-name
```

job-name is the name of the job you wish to notify. For example:

```
@@NOTIFY=JOB1
```

As with messages sent via the AMOS SEND command, remember that the specified job must be SET NOGUARD in order to receive the message.

On computers configured with AlphaNET, you can specify the computer on which the job is located, either by including the numeric system ID, or by using an equivalent ersatz name. For example:

```
@@NOTIFY=1694695430-JOB3
```

or

```
@@NOTIFY=ACCNTG:JOB6
```

Remember that you must specify the system ID from the viewpoint of the computer running FAXSER, not the viewpoint of the person sending the fax.

Sending Multiple Faxes From One Text File

Sometimes it is useful to be able to create one text file which AlphaFAX will send as multiple faxes. A typical example is a program which creates customized faxes to go to a list of customers. Since each fax must be unique, a distribution list won't do the job. It would be possible to create separate files for each fax, but that makes the task more complicated than it needs to be.

AlphaFAX provides for such cases with the @@NEWDOCUMENT command. When AlphaFAX sees one of these commands, it treats it as the end of the current fax and the beginning of a new one, just as if it was in a separate file. by placing @@NEWDOCUMENT commands within a file, you can use a single text file to send as many faxes as you wish.

Remember that the @@NEWDOCUMENT command behaves exactly like a new file is being processed: none of the settings from the previous fax carry over to the new fax.

A file making use of @@NEWDOCUMENT might look like:

```
@@FINE
@@TELNO LGB
This text will be sent to LGB.
@@NEWDOCUMENT
@@TELNO 1 714 957-8705
This text will be sent to 714 957 8705, but not in
fine mode.
```

CHAPTER 5

LOOKING AT FAX ACTIVITY

Looking at the Active Fax Queue

AlphaFAX maintains a queue of faxes that are waiting to be sent, printed, previewed, or deleted. It processes each item in the queue in turn. To display the current contents of the fax queue, you can use the FAXQUE command. For example:

FAXQUE RETURN

A sample queue display looks like this:

```
 Fax ID       Request     Retries  Schedule   Telno

PO 3124       Send           0     Active    LGB
SALLY         Send           2     12:23:45  1 714 957-8705
2M1401        Print          0     Next
              New request    0     Next

The AlphaFAX server is currently sending
```

Note that the queue contains entries for various types of activities and also shows when a request is scheduled to be operated on.

Daily Fax Activity Log

AlphaFAX maintains transaction log files for all faxes sent and received. These are stored as standard ASCII text files. Separate log files are maintained for incoming and outgoing faxes, with a new log file being started each day.

All fax logs are in the ersatz device FAXLOG:. They are given names of yymmdd.IN for incoming fax logs, and yymmdd.OUT for outgoing fax logs (*yy* = last two digits of the year, *mm* = two-digit month of the year, and *dd* = two-digit day of the month). For both outgoing and incoming faxes, the filename yymmdd corresponds to the date to which the log applies. For example, the log for a fax sent on February 21, 1990 has the name 900221.OUT.

Displaying a Fax Activity Log

You can look at each log file directly by using the AMOS TYPE command to display it on your terminal. (You can also print the file.) For example:

TYPE 900215.OUT RETURN

A sample fax log looks like this:

```
                    MESSAGES SENT REPORT

                      Banzai Institute
                       9 January 1990

Start    Elapsed        Partner                    Pages    Result       ID

09:13     00’39    MicroMark                          0    No Answer  PO 2422
09:18     02’47    MicroMark                          2    OK         PO 2422
12:09     06’35    32 2 2417139                       3    OK         jp
15:53     01’29    2706322                            1    OK         RBC
19:09     01’47    1 607 279 6252                     2    OK         0J1902
```

Using FAXSTS

If you just want to look at today’s log of outgoing faxes, or a log for a particular day, instead of using the TYPE command, use the AlphaFAX FAXSTS command. The format of the FAXSTS command to see today’s log is:

FAXSTS

If no log has been created yet today, FAXSTS tells you so. If you want to see a prior day’s outgoing fax log, use FAXSTS in the following format:

FAXSTS n

n is the number of days prior to today for which you wish to see a log. For example, to see the log for the day before yesterday, type at AMOS command level:

FAXSTS 2 RETURN

Deleting a Fax Log

To delete a fax activity log, just use the AMOS ERASE command. For example:

ERASE 901123.IN RETURN

CHAPTER 6

CONVERTING RECEIVED FAXES TO BMP FILES

BMP Files

AlphaFAX can include images in outgoing faxes, providing that those images are in the BMP image format. This format is used by AMIGOS (the Alpha Micro Graphics Operating System) as a standard method of storing and displaying bitmap images. This format is different from the format used to store received fax documents.

Converting Fax Documents to BMP Files

You can convert a received fax document (.FAX extension) to a BMP file (.BMP extension) by using the FAX2BM program. This program converts the entire fax document, including headers and blank space, into a BMP format file. Because the compression method used for fax documents has been optimized for that purpose, BMP files are typically considerably larger than the original fax document file.

Using FAX2BM

If you want to convert a fax document to a BMP file, use the FAX2BM command in the following format:

```
FAX2BM fax-name
```

fax-name is the name of the fax document you wish to convert. For example:

FAX2BM 0Y1700 RETURN

The resulting BMP file has the same name as the fax document, but an extension of .BMP.

CHAPTER 7

USING ALPHAFAX WITH ALPHAWRITE

AlphaWRITE Versions

When used with AlphaWRITE 2.1 or later, AlphaFAX offers support for a variety of proportionally spaced fonts in several different sizes, in addition to its basic monospaced font. This can give your faxes a highly polished look approaching laser printer quality. When used with AlphaWRITE 1.2A or AlphaWRITE 2.0, AlphaFAX is limited to basic monospaced type.

AlphaWRITE Header Settings

For best results, be sure to set up the header of your AlphaWRITE document correctly.

For AlphaWRITE 1.2A, Set:

Page Length: 66
Text Length: 57
Space Before Footnote: 0

and, remember to use the PAGINATE command to format the fax into pages of the correct length.

For AlphaWRITE 2.0 and Later, Set:

Top Margin: 0
Title Length: 0
Page Length: 11"
Text Length: 9"
Space Before Footnote: 0
Footer Length: 0

It is very important to set the top margin and title length to zero, or you will not be able to select fine mode transmission.

General Hints

Faxing documents with AlphaWRITE is just like printing a document with AlphaWRITE. Just be sure to remember the following points:

- Make sure you have a hard carriage return after all @@ commands. Without one, AlphaWRITE will reformat the text and prevent the command from being passed to AlphaFAX correctly.
- Remember to make the @@FINE command the very first line in the file— even before any blank lines— or it will be ignored.
- So that AlphaWRITE knows how to paginate your document correctly, try to use the @@OVERLAY command to output all images, instead of the @@IMAGE command.

Proportional Text

AlphaWRITE 2.1 and later supports the use of proportional text within fax documents. Proportional text is text where each character is a different width, like the text of this sentence, as opposed to monospaced text where each character is the same width, as with a traditional typewriter.

Select proportional text with AlphaFAX just as you do for any other printer. That is, within the header or margin block, select proportional spacing. Select the typeface and typesize using the STYLE key.

AlphaFAX offers several typefaces in a number of point sizes. The fonts supplied with AlphaFAX are shown below:

TYPEFACE	CODE	SIZES	STYLES
Helvetica	H	4, 5, 6, 7, 9, 12	Normal, Bold, Italic
Times Roman	T	4, 5, 6, 7, 9, 12	Normal, Bold, Italic
Bitstream Charter	B	4, 5, 6, 7, 9, 12	Normal, Bold, Italic

Not all of the fonts and sizes listed above may be installed on your computer. Check with your system operator to determine which fonts are available.

For general faxing Bitstream Charter is probably the best typeface to use, as it reproduces well at the relatively low resolution of fax technology.

When using proportional typefaces, be sure to include a @@FINE command to select fine mode transmission. The quality of the proportional typefaces suffers greatly when used in standard mode.

CHAPTER 8

INTERFACING ALPHAFAX TO YOUR PROGRAMS

AlphaFAX and Your Programs

In addition to creating and sending faxes using word processing programs, AlphaFAX also excels at working with your other programs. Because AlphaFAX looks just like another printer on your computer, any report you currently print can be sent via fax. By making simple modifications to existing programs, you can send the information directly to its final destination. Typical applications include:

- Automatically generate invoices and send them by fax, eliminating the time spent in the mail, and getting payment to you sooner.
- Send automatic order confirmations as soon as the order has been entered.
- Send purchase orders directly to suppliers via fax.
- Send fax newsletters to keep your clients up to date on your business.
- Transmit transcripts and other records directly from your database to a fax machine.
- Send product information as soon as it is requested, rather than having to mail it out.

Doing any of these is simply a matter of including the outgoing fax telephone number into an existing report. Some tips to keep in mind are:

- In addition to a voice phone number, keep a fax number in your customer and vendor databases.
- You can generate a single report containing faxes to many different people by using the @@NEWDOCUMENT command.
- Take advantage of the @@IMAGE and @@OVERLAY commands to make your fax transmissions look just like a hardcopy version by including logos, signatures, and pre-printed forms.
- Use monospaced text for sending existing 80-column reports.

- Use compressed print for sending existing 132-column reports.
- Make use of the other embedded formatting codes to produce professional looking documents. (See Appendix B for formatting codes.)
- Use distribution lists for mass faxings of the same document.

APPENDIX A

SAMPLE FAX

The following sample may be used as a prototype document:

```
@@TELNO=9578705
@@ID=your name
@@IMAGE=IMG:FAXCOV;2.4
   TO:

 ATTN:

 FROM:

PAGES: 1 (including cover)

_____________________________________________________

This is sample text.
```

That file produces the following fax on the destination fax machine:

Wed 25 Mar 1992 10:37 BanZai Institude Page 01

FAX Transmission
Transmitted via AlphaFAX

In case of error please call for retransmission

TO:

ATTN:

FROM:

PAGES: 1 (Including cover)

This is a sample text

APPENDIX B

DOCUMENT FORMATTING CODES

Document Formatting Codes

When using AlphaFAX with a word processor like AlphaWRITE, the word processor provides access to the special formatting features of AlphaFAX such as underlining, bold text, and italic text. Use the word processor's features to access these just as you do with any other printer.

When using AlphaFAX without a word processor, such as creating special application software that directly sends reports via fax, you need to use special embedded codes to access these features. This appendix defines those codes.

General Command Format

All embedded formatting codes have the same general format:

```
<esc> length command
```

esc is the escape character (ASCII 27 decimal, 33 octal, 1B hex), followed by a ***length*** byte which contains the length of the following ***command*** string in bytes.

The spaces and angle brackets in the formatting code commands shown in this appendix have been included to improve readability. Don't include spaces or brackets in your formatting commands.

Most AlphaFAX commands have a length of 2. This corresponds to sending a decimal 2 or control-B character for the length field.

Note that ***esc*** and ***length*** are actual binary data, not ASCII text characters. (The binary value 2 and the ASCII text character "2" are not the same.) Here are two examples of ways to embed the underscored text command (described more fully later in this appendix):

```
<esc> <decimal 2> f u
```

From Within AlphaVUE

Since the AlphaVUE text editor allows you to enter binary data (for example, control-characters), you can type the embedded data directly into your text file. You signal AlphaVUE you are going to enter binary data by first typing a Control-G; then you press the key(s) corresponding to the binary data you want to enter.

The slash between two keys in the sample below indicate that you hold down the first key while you press the second key. For example, to type a binary 2 into the file, type CTRL/G (hold down the CTRL key while you press the G key) followed by CTRL/B (hold down the CTRL key while you press the B key). A Control-B is the binary value 2.

To enter the underscored text formatting command show above, type the following in AlphaVUE:

CTRL/G ESC CTRL/G CTRL/B **fu**

By Way of an AlphaBASIC Program

Your application program can also insert the formatting codes into a text file. For example, here is an AlphaBASIC program that opens a text file and inserts the underscored text formatting command string shown above.

```
OPEN #1,"TEST.TXT",OUTPUT
! Write ESC, binary 2, string "fu" to the file.
PRINT #1 CHR$(27);CHR$(2);"fu";
CLOSE #1
```

Selecting Underscored Text

To select underscored text, use the two-byte `fu` command. The full command sequence is:

```
<esc> <decimal 2> f u
```

To turn underscoring off again, repeat the command. Each occurrence of the `fu` command toggles underscoring on or off.

Selecting Bold Text

To select bold text, use the two-byte `fb` command. The full command sequence is:

```
<esc> <decimal 2> f b
```

To turn bold text off again, repeat the command. Each occurrence of the `fb` command toggles bold text on or off.

Selecting Italic Text

To select italic text, use the two-byte `fi` command. The full command sequence is:

```
<esc> <decimal 2> f i
```

To turn italic text off again, repeat the command. Each occurrence of the `fi` command toggles italic text on or off.

Selecting Compressed Print

Compressed print reduces the width of each character so that you can place wider reports, such as those designed for 132 columns, on a single page. To select compressed print, use the two-byte `fc` command. The full command sequence is:

```
<esc> <decimal 2> f c
```

To turn compressed print off again, repeat the command. Each occurrence of the `fc` command toggles compressed print on or off.

Selecting Proportional Spacing

To select proportional text, use the two-byte `fp` command. The full command sequence is:

```
<esc> <decimal 2> f p
```

To turn proportional off again, repeat the command. Each occurrence of the `fp` command toggles proportional text on or off.

You may also select proportional printing, regardless of the current setting of the proportional flag, using the two-byte `sp` command. The full command sequence is:

```
<esc> <decimal 2> s p
```

To force proportional text off, regardless of its current state, use the two-byte `sf` command. The full command sequence is:

```
<esc> <decimal 2> s f
```

Selecting a Proportional Font

To select a specific proportional font, use the `ff` command. Within this command, specify the name of the font you wish to select. The full command sequence is:

```
<esc> <length> f f fontname
```

fontname is the name of the selected font, taken from the table below; and ***length*** is the length of the font name text string plus two.

FONT	FONTNAME	LENGTH
Bitstream Charter	bitstream_charter	19
Helvetica	helvetica	11
Times Roman	times_roman	13

For example, to select the Helvetica font, send the command stream:

```
<esc> <decimal 11> f f h e l v e t i c a
```

Choosing Font Size

To select the desired size for a proportional font, use the three-byte `fs` command. The full command sequence is:

```
<esc> <decimal 3> f s pointsize
```

pointsize is the desired font size, in points, encoded as the size in points plus 31 decimal. For example, to select 12 point type, use the following command:

```
<esc> <decimal 3> f s <decimal 43>
```

The font size command only affects proportional text. Remember that each proportional font is only available in a fixed number of sizes. Selecting the wrong size leads to unpredictable results in the fax output.

Moving to an Absolute Position

To move to an absolute print position on the current line, use the five-byte `pa` command. The full command sequence is:

```
<esc> <decimal 5> p a position
```

position is an encoded, three-byte sequence specifying the position to move to in 1/200" units.

The position argument is an encoded 12-bit number, sent as three bytes. The first byte contains the upper four bits of the 12-bit number, with a bias of 32 decimal added in. The second byte contains the middle four bits, with a bias of 32 decimal. The third byte contains the low-order four bits, with a bias of 32 decimal.

Moving to a Relative Position

To move to a print position on the current line, relative to the current print position, use the five-byte `pr` command. The full command sequence is:

```
<esc> <decimal 5> p r position
```

position is an encoded, three-byte sequence specifying the position to move to in 1/200" units.

The position argument is an encoded 12-bit number, sent as three bytes. The first byte contains the upper four bits of the 12-bit number, with a bias of 32 decimal

added in. The second byte contains the middle four bits, with a bias of 32 decimal. The third byte contains the low-order four bits, with a bias of 32 decimal.

Resetting Text Attributes

You can reset all text attributes, including bold, underscore, italic, compressed, and proportional, with a single two-byte `fr` command. The full command sequence is:

```
<esc> <decimal 2> f r
```

This command is often useful at the beginning of a new section of a document to make sure that no settings are left over from previous sections.

Combining Text Effects

You can combine special text effects by combining the embedded commands. Following one command with another combines the text effects to produce combinations such as bold underscored text.

APPENDIX C

USING THE MASTER FAX LOG

The Master Fax Log

In addition to the fax activity logs described in Chapter 5, AlphaFAX can also create an optional master fax log. By placing a MASTERLOG=TRUE statement in the FAX.INI file, master logs will be created for all incoming and outgoing faxes. These log files have been specifically designed for processing by other programs to analyze fax usage and estimate costing, and to be sorted in a whatever order you most prefer.

AlphaFAX maintains two master fax logs: one for outgoing faxes and one for incoming faxes. The outgoing log is stored as FAXLOG:FAXLOG.OUT, while the incoming log is stored as FAXLOG:FAXLOG.IN. Both files share the same fixed format. This format has been designed to make it easy for programs to process the data. The format of each line in the master log is as follows:

COLUMN	CONTENTS
1-4	Year of fax transmission
6-7	Month of fax transmission, in the range 1-12.
9-10	Day of fax transmission
12-16	Time the fax transmission commenced
18-22	Elapsed time of fax transmission
24-33	The fax ID, as supplied in a @@ID command or generated by AlphaFAX
35-64	The telephone number, as dialed
66-95	The ID returned by the remote fax machine
97-99	The page count
101-110	The result code
112-116	Transmission speed
118	Transmission mode (N = normal, F = fine)

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